Junwei Zhang

Computer Science Undergraduate

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Google scholar

Education

2023.09–Present **B.S. in Computer Science**, *University of Chinese Academy of Sciences, School of Computer Science and Technology*, Beijing, China
- GPA: 3.97/4.0
- Rank: 1/86

Research Interests

LLM Hallucination, Agent Memory and Recursive Self-Improvement.

Research Experiences

2026.01–2026.10 **Hallucination Mitigation via Abstention**, *ICT, CAS*, Beijing,
Advisor: Prof. Fei Sun
- Revisit abstention training and find its advantage stems from better answering;
- Develop a simple and more controllable approach that outperforms abstention training.

2026.04-2026.05 **Improve LLM Knowledge Recall via Reinforcement Learning**, *ICT, CAS*, Beijing,
Advisor: Prof. Fei Sun
- Discover that RL significantly improves direct knowledge recall in LLMs;
- Derive an empirical explanation for this phenomenon.

2025.12–2026.01 **Confidence Calibration in Large Language Models**, *ICT, CAS*, Beijing,
Advisor: Prof. Fei Sun
- Propose a generalizable unsupervised method for confidence calibration in LLMs;
- Achieve state-of-the-art performance among unsupervised calibration baselines.

Research Internships

2026.02–Present **Research Intern**, *Recursive Self-Improvement Team, Meituan Inc.*, Beijing,
Mentors: Dr. Jingang Wang & Dr. Xin Chen
Investigate LLM hallucination mitigation and agent experiential learning.

Preprints & Publications

arXiv 2026 **Junwei Zhang**[*], Yifei Wang[*], Wanli Yang, Hexiang Tan, Rui Tang, Xin Chen, Jingang Wang, Du Su, Fei Sun. *Strong yet Controllable LLM Abstention without Learning to Abstain*. Under review at ICLR 2027.
[*]Equal contribution.

arXiv 2026 Wanli Yang, Hongyu Zang, **Junwei Zhang**, Wenjie Shi, Du Su, Jingang Wang, Xueqi Cheng, and Fei Sun. *Beyond Reasoning: Reinforcement Learning Unlocks Parametric Knowledge in LLMs*.

ACL 2026 Hexiang Tan, Wanli Yang, **Junwei Zhang**, Xin Chen, Rui Tang, Du Su, Jingang Wang, Yuanzhuo Wang, Fei Sun, and Xueqi Cheng. *BaseCal: Unsupervised Confidence Calibration via Base Model Signals*.

Projects

2024.09–2025.09 **YSYX-project (One student One Chip Project)**

- Design and optimize a RISC-V processor using Chisel HDL;
- Implement a RISC-V virtual machine to perform differential testing;
- Develop lightweight runtime environment to support application-level workloads.

Honors & Awards

2024, 2025 **National Scholarship (Top 2%)**
Ministry of Education of P.R. China

2025 **Finalist award in Interdisciplinary Contest In Modeling (Top 3%)**
Consortium for Mathematics and its Applications

2024, 2025 **First Class Academic Scholarship (Top 5%)**
University of Chinese Academy of Sciences

2024, 2025, 2026 **Triple-A student**
University of Chinese Academy of Sciences

Skills

Programming Python, PyTorch, C, Verilog, Chisel/Scala

Development Git, Neovim, Linux

Languages Chinese (Native), English (Fluent)